

TECHNOLOGICAL UNIVERSITY (THANLYIN)
DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

PROJECT MANAGEMENT SYSTEM

BY
MA KYAW KAY ZIN HAN
VI IT - 6

GRADUATION THESIS

OCTOBER, 2025
THANLYIN

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A THESIS
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TECHNOLOGICAL UNIVERSITY (THANLYIN)
DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

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ABSTRACT

This thesis focuses on the development of a Project Management System (PMS) that supports the efficient planning, tracking, and evaluation of multiple projects across various departments. The system enables users to create projects, assign tasks to employees, track progress, and manage departmental responsibilities. Key features include task management, department and employee linkage, automatic timeline calculation, and Gantt chart visualization. It also provides performance measurement tools such as Planned Value (PV) and Actual Cost (AC) to assess budget and time efficiency. The PMS is designed to be flexible and applicable not only to IT projects but also to other fields like construction and education. It helps project managers make informed decisions using performance data and visual insights such as bar charts and donut charts.

In this thesis, algorithms and tools are applied to enhance both functionality and security. Timeline calculation and Gantt chart rendering are handled using custom time-based algorithms, while Earned Value Management (EVM) metrics such as Earned Value (EV), Cost Variance (CV), Schedule Variance (SV), Cost Performance Index (CPI) and Schedule Performance Index (SPI) are automatically computed to monitor project performance. Status transitions and task progress are managed with a workflow state algorithm that optionally calculates task completion percentages. For security, Secure Hash Algorithm (SHA-256) is used to hash user passwords securely before storing them in the database. Flutter, Dart, Visual Studio Code and SQLite are used as the main development tools to build a responsive, database-driven application.

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